

DrawMate

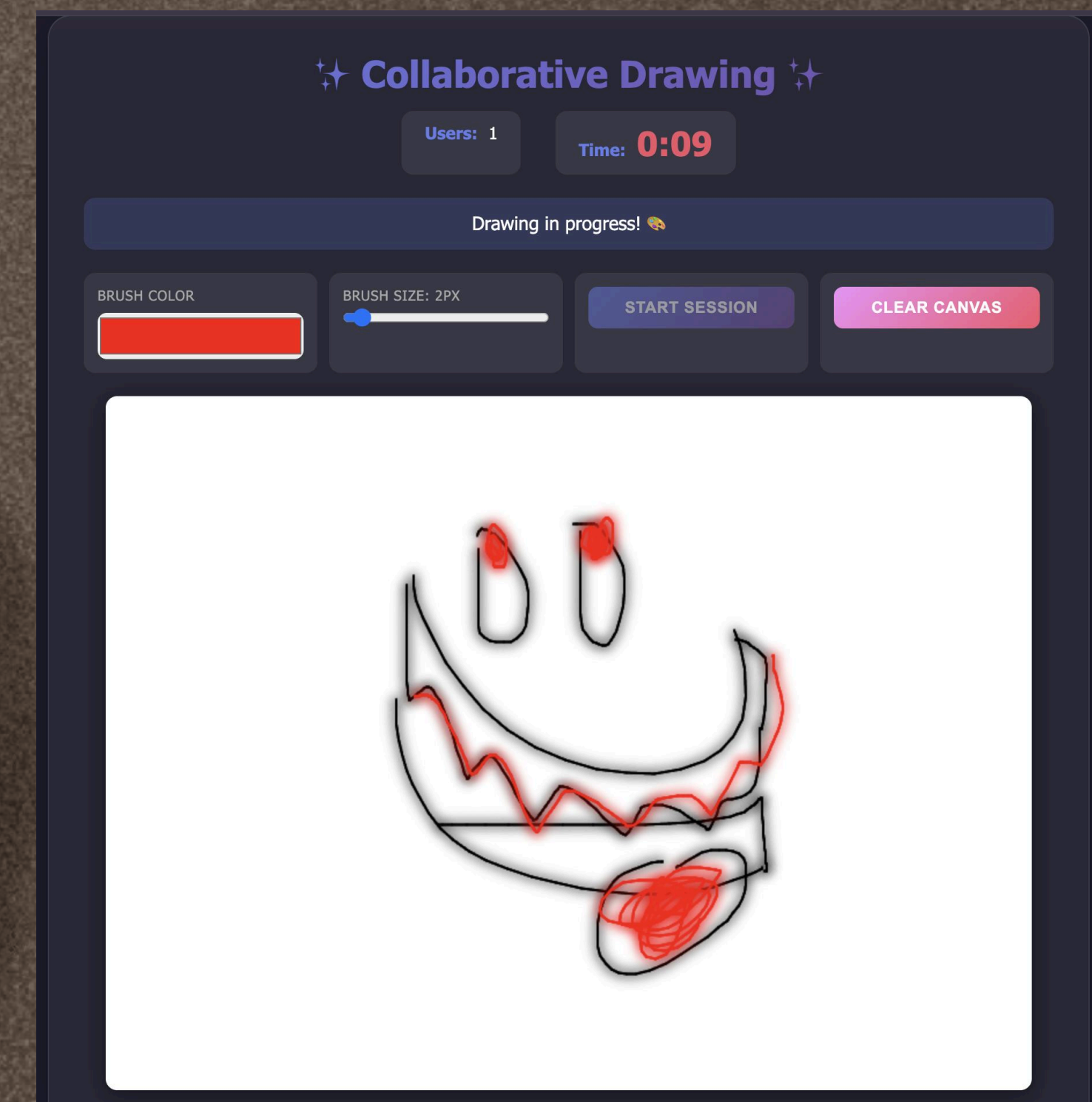
PIX Dec 2025 ▪ Uma Dhamija

Project Overview

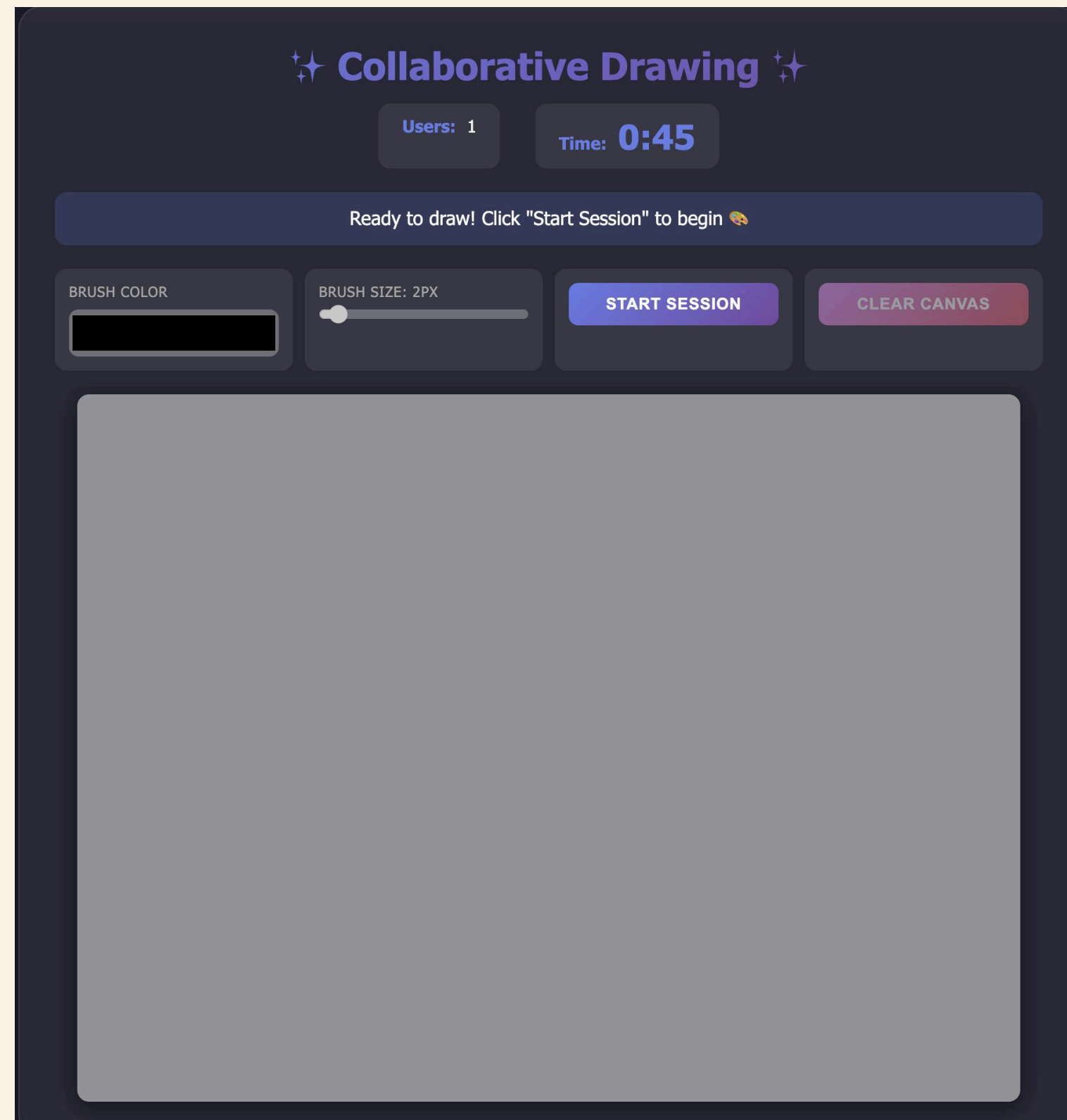
01

DrawMate is a real-time collaborative drawing app where multiple people can draw together on the same canvas during timed 45-second sessions.

It's basically a shared whiteboard that works over the internet – think Google Docs but for drawing, with the added excitement of racing against the clock.



Project Overview



Who it's for: Anyone who wants to sketch ideas together remotely, play timed drawing games like Pictionary with friends, collaborate on quick visual brainstorms, or just mess around making art together under time pressure.

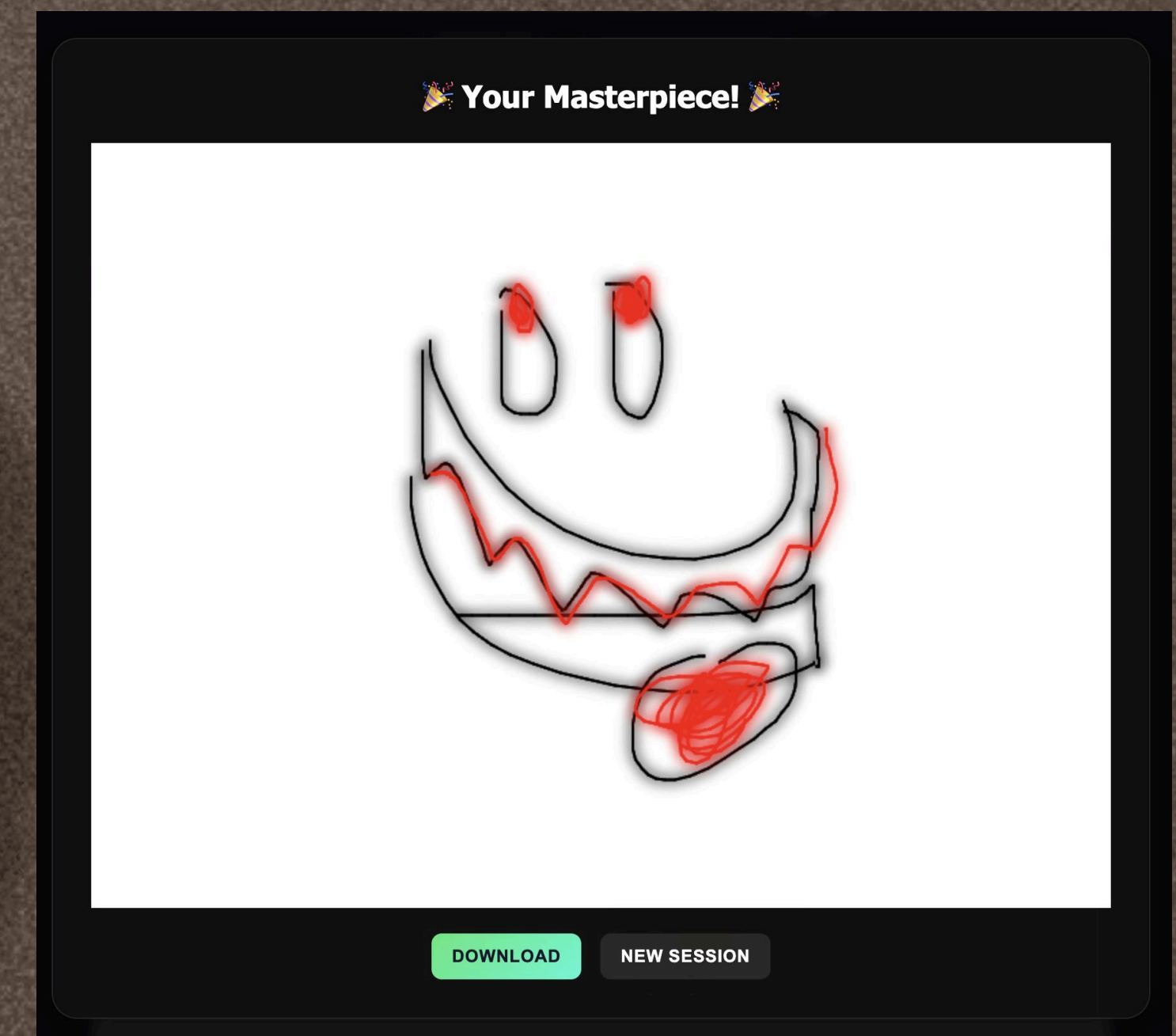
How it works: You open it in your browser and you're connected to a shared canvas. Anyone can start a 45-second drawing session by clicking "Start Drawing Session." Once started, everyone draws together in real-time – when one person draws a line, everyone else sees it appear instantly. You can pick your own color and brush size, and anyone can clear the canvas during the session to start fresh. When the timer hits zero, the session ends and a modal pops up showing everyone the completed collaborative artwork, which you can download as a PNG.

Process

01

I started with the planning document where I sketched out the basic idea and wrote pseudocode. That helped me think through the client-server communication before diving into actual code.

I decided to build the server first because it felt like the foundation – once I got the server broadcasting messages correctly, I focused on the client-side drawing logic without stressing about networking.



Code

The app uses a client-server architecture with Socket.io handling real-time communication. The server acts as both a message relay and the authoritative timer – it manages the 45-second countdown and broadcasts time updates to all clients. The clients (browsers) handle the canvas rendering and mouse input, but can only draw during active sessions controlled by the server.

```
1  PSUEDOCODE
2
3  // Shared data that lives on the server
4  type Stroke = {
5    points: { x: number; y: number }[];
6    color: string;
7    size: number;
8  }
9
10 // Server
11 on client connect:
12   broadcast new user count to everyone
13
14 on "stroke" message from any client:
15   broadcast the exact stroke data to ALL clients (including sender)
16
17 on "clear" message from any client:
18   broadcast "clear" command to ALL clients
19
20 on client disconnect:
21   broadcast updated user count
22
23 // Client (TypeScript in browser)
24 connect to Socket.IO server
25 maintain local drawing state:
26   isDrawing, currentStroke[], currentColor, currentSize
27
28 on mouse down:
29   start new stroke, record first point
30
31 on mouse move while drawing:
32   add point to currentStroke
33   draw line segment locally (instant feedback)
34
35 [and more]
36
37 UI Controls:
38   • Color picker → instantly changes currentColor
39   • Size slider (1-20px) → instantly changes currentSize + live display
40   • Clear button → emits "clear" event
```

✦ Collaborative Drawing ✦

Users: 2

Time: 0:45

Ready to draw! Click "Start Session" to begin 🎨

BRUSH COLOR

BRUSH SIZE: 2PX

START SESSION

CLEAR CANVAS

Thank you

Especially to Rabiya, Yukti, Priya, and Jason